

IBM BOB 2.0 HACKATHON · SEPTEMBER 2026

FlakeHunter

Find, fix and prove flaky tests,
with IBM Bob agents working in parallel.

flake-hunter.onrender.com · github.com/antthein/Flake_Hunter

THE PROBLEM

Your CI is red. Your code is fine.
Your tests are flaky.

15%

of CI runs were fully green on our demo app,
with **zero code changes**.

Hours lost clicking “re-run” and waiting.

Merges blocked by failures that aren’t real.

Trust lost: red builds get ignored, so real bugs
slip through.

WHY IT HAPPENS

Five causes behind most flaky tests

Cause	What happens	Example
Timing	A wait is too short on a busy machine	<code>sleep(1)</code> for work that takes 1.2 s
Test order	A test passes only if another ran first	B relies on A's leftover setup
Shared state	Tests change the same global, file or row	a module-level dict never reset
Randomness	An unseeded value is sometimes “unlucky”	<code>random.choice</code> picks an invalid item
Network	A real service is slow or down	a payment API times out

THE SOLUTION

One command: find, fix and prove

01

Detect

Run the suite N times in random order. Pass and fail on the same code means flaky.

02

Hint

An AST scan adds clues: timing, random, network, shared state.

03 • IBM BOB

Diagnose

One Bob agent per flaky test, in parallel, each in an isolated copy.

04

Verify

Apply the fix, re-run 50× in random order. Kept only at 50/50.

05

Report

Before → after, root causes, Bob's explanations, diffs, CI time saved.

```
$ flakehunter run ./my-project
```

IBM BOB AS THE ENGINE

Four Bob agents, side by side

4 flaky tests

with pass/fail rates and hints
from detection



bob run · agent 1 · copy 1

bob run · agent 2 · copy 2

bob run · agent 3 · copy 3

bob run · agent 4 · copy 4



FlakeHunter checks

only the test file changed? →
diff → apply → 50/50 verify

Each agent reads its task from FLAKEHUNTER_TASK.md, fixes only its test file, and replies in JSON with the root cause and a two-sentence explanation.

SAFETY BY DESIGN

Nothing is kept unless it is proven

Isolated

Bob works only in a temporary copy, never in the real project.

Scoped

If Bob edits any file other than the test file, the fix is rejected.

Proven

A fix is kept only if its test file passes 50 of 50 random-order runs.

Reversible

Otherwise the original file is restored, byte for byte, automatically.

RESULT • REAL IBM BOB RUN

From flaky to green in one run

15% → 100%

CI runs fully green

4 → 0

flaky tests

4 / 4

Bob fixes kept, each verified 50/50

≈ 25 h

CI time saved per month*

Demo shop app, 15 tests, 26 Sep 2026. *Assumes 10 CI runs a day and 5 minutes per re-run.

WHAT BOB FOUND

Four root causes, four minimal fixes

Flaky test	Root cause Bob found	Bob's fix	Verified
Inventory starts empty	Shared dict leaks between tests, depending on order	setup_function resets it	50/50
Order within SLA	Random sleep up to 0.15 s breaks a 0.1 s limit	monkeypatch the sleep length	50/50
Payment returns OK	Gateway call times out 30% of the time	monkeypatch the timeout source	50/50
Promo winner eligible	Unseeded random choice can pick an ineligible person	monkeypatch random.choice	50/50

Every fix changed only the test file, using standard pytest tools. Bob also explained each cause in two sentences for the report.

IBM BOB AS THE BUILDER

Built step by step with Bob 2.0

Plan mode

Bob asked scoping questions and wrote PLAN.md. Two review rounds fixed the design.

Agent mode

A fresh task per step, each starting from PLAN.md: demo app, runner, scanner, CLI, diagnoser, patcher, reporter.

Review loop

Every step tested before the next. Bugs found and fixed by Bob, such as Windows cutting off prompts.

Bob Shell

The finished tool calls bob run headless, in parallel, as its fixing engine.

After our Bob credits ran out, the web UI, Dockerfile and README were added with another AI assistant. Full history is in git.

HOW TEAMS USE IT

Install once, run it where tests run

On a laptop

```
pip install git+https://github.com/antthein/Flake_Hunter.git  
flakehunter run .
```

In CI, nightly

Find new flaky tests, let Bob fix them, open a pull request with only the verified fixes.
Next step

Team dashboard

flakehunter serve: watch detection, Bob's fixes and verification live, then review the report.

Roadmap: GitHub Action, PyPI package, Jest and JUnit support.

TRY IT

**Press one button.
Watch the tests turn green.**

flake-hunter.onrender.com

github.com/antthein/Flake_Hunter

The live demo runs detection and verification for real and replays our recorded IBM Bob run. The first visit may take about a minute to wake the free server.